

DATA SCIENCE RESOURCES

Data Visualization Projects

Project 1: Sales Dashboard in Tableau

Problem Statement: Sales leaders need real-time visibility into regional, category-wise, and monthly sales to drive revenue and decision-making.

Objective: Create an interactive Tableau dashboard to monitor sales performance, trends, and KPIs like revenue, profit margin, and sales growth.

Dataset Source: [Superstore Sales Dataset \(Kaggle / Tableau Public\)](#)

Key Features

- Filters for category, region, and date
- KPIs: Total Sales, Profit, Quantity, Discount
- Line chart for trend analysis
- Geo map for state-wise sales
- Bar chart for top-performing products

Tools Used: Tableau Desktop or Tableau Public

Outcome: An executive-ready dashboard for identifying underperforming regions, forecasting growth, and making data-driven decisions.

Project 2: Indian Startup Funding Analysis

Problem Statement: Investors and analysts want to understand startup trends in India: funding size, sector growth, and location-wise investment patterns.

Objective: Create a dashboard to visualize startup funding trends in India across sectors, cities, and time.

Dataset Source: [Indian Startup Funding Dataset](#) (Kaggle)

Key Features

- Total Funding Raised by Year
- Sector-wise and City-wise Investment Distribution
- Funding Trends Over Time
- Top Funded Startups

Tools Used: Tableau or Power BI, Excel, Python (optional preprocessing)

Outcome: A complete visual analytics dashboard to explore startup ecosystem insights, ideal for VCs, analysts, or journalists.

Project 3 : YouTube Trending Video Analytics

Problem Statement: Content creators and digital marketers want to decode trends and audience behavior from YouTube's trending videos.

Objective: Analyze YouTube trending video datasets to find common traits in viral content—views, tags, likes/dislikes ratio, category performance.

Dataset Source: [YouTube Trending Videos Dataset](#) (Kaggle)

Key Features:

- Views vs. Likes ratio
- Trending Video Category Breakdown
- Word Clouds for Common Tags
- Publishing Time Analysis (hour/day of week)

Tools Used: Tableau, Power BI, or Python (Seaborn + Plotly Dash for interactive dashboards)

Outcome: A dashboard revealing what makes videos trend—valuable for content strategy, SEO, and YouTube campaign planning.

Project 4 :IPL Scorecard Analysis Using Power BI

Problem Statement: Cricket enthusiasts and sports analysts need deeper insights into player performance, team stats, and match outcomes across IPL seasons.

Objective: Build an IPL dashboard in Power BI showing season stats, batting and bowling performance, win/loss ratios, and player comparisons.

Dataset Source: [IPL Matches & Deliveries Dataset](#) (Kaggle)

Key Features

- Player Run/Strike Rate Charts
- Team Win Ratios by Season
- Toss Impact vs. Match Result
- Most Runs/Wickets Leaderboard
- Filters: team, player, season

Tools Used: Power BI Desktop, DAX, Power Query

Outcome: An interactive cricket analytics dashboard suitable for presentation to media, fans, or cricket boards.

Project 5 : Indian Census Visualization

Problem Statement: Policy-makers and researchers need a simple way to understand population demographics, literacy rates, and gender distribution across Indian states.

Objective: Visualize key census indicators using charts, maps, and filters for state/district-level data.

Dataset Source: [India Census 2011 Dataset](https://data.gov.in) (data.gov.in)

Key Features

- State-wise Population Pyramid
- Literacy Rate Heatmap
- Gender Ratio by District
- Urban vs. Rural Population Distribution

Tools Used: Tableau or Power BI with shapefiles for map visualizations

Outcome: An informative dashboard that supports demographic research, policy making, and public data storytelling.
